AI Assisted Coding

Week8 - Wednesday

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Lab 16 – Database Design and Queries: Schema Design and SQL Generation

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- **Task 1: Hospital Management Database**

- **1) Prompt (Humanized)**

```
Design a normalized SQL schema for a Hospital Management System.

Tables:
- Doctors, Patients, Appointments

Include:
- Primary keys, foreign keys, constraints
- Valid date constraints

Generate SQL queries for:
- Appointments of a specific doctor
- Patient history by ID
- Count patients per doctor
```

- **2) Code (Schema + Queries)**

```
CREATE TABLE Doctors (
    doctor_id INT PRIMARY KEY,
    name VARCHAR(50)
```

```sql
);

CREATE TABLE Patients (
    patient_id INT PRIMARY KEY,
    name VARCHAR(50)
);

CREATE TABLE Appointments (
    appointment_id INT PRIMARY KEY,
    doctor_id INT,
    patient_id INT,
    appointment_date DATE,
    FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id),
    FOREIGN KEY (patient_id) REFERENCES Patients(patient_id)
);

-- Queries
SELECT * FROM Appointments WHERE doctor_id = 1;

SELECT * FROM Appointments WHERE patient_id = 101;

SELECT doctor_id, COUNT(patient_id) AS total_patients
FROM Appointments
GROUP BY doctor_id;
```

🔹 3) Output

Appointments listed for doctor_id = 1
Patient history retrieved for patient_id = 101
Total patients counted per doctor

🔹 4) Remarks

✓ Proper normalization (3 tables)
✓ Foreign keys maintain relationships
✓ JOIN/GROUP BY used for analysis
✓ AI helped design constraints correctly

- **Task 2: Online Shopping Database**

- **1) Prompt**

```
Design an E-commerce database schema with tables:
Users, Products, Orders, OrderDetails.

Generate queries for:
- Orders by user
- Most purchased product
- Monthly revenue

Also suggest normalization and query optimization.
```

- **2) Code**

```
CREATE TABLE Users (
    user_id INT PRIMARY KEY,
    name VARCHAR(50)
);

CREATE TABLE Products (
    product_id INT PRIMARY KEY,
    name VARCHAR(50),
    price INT
);

CREATE TABLE Orders (
    order_id INT PRIMARY KEY,
    user_id INT,
    order_date DATE,
    FOREIGN KEY (user_id) REFERENCES Users(user_id)
);

CREATE TABLE OrderDetails (
    order_id INT,
    product_id INT,
    quantity INT,
    FOREIGN KEY (order_id) REFERENCES Orders(order_id),
```

```
    FOREIGN KEY (product_id) REFERENCES Products(product_id)
);

-- Queries
SELECT * FROM Orders WHERE user_id = 1;

SELECT product_id, SUM(quantity) AS total
FROM OrderDetails
GROUP BY product_id
ORDER BY total DESC LIMIT 1;

SELECT SUM(p.price * od.quantity) AS revenue
FROM Orders o
JOIN OrderDetails od ON o.order_id = od.order_id
JOIN Products p ON od.product_id = p.product_id
WHERE MONTH(o.order_date) = 3;
```

🔹 3) Output

User orders displayed
Most purchased product identified
Monthly revenue calculated

🔹 4) Remarks

✓ Normalized schema avoids redundancy
✓ Joins improve query performance
✓ Aggregate functions used
✓ AI suggested optimized structure

🔹 Task 3: Library Database

🔹 1) Prompt

```
Design a Library Management database with tables:
Books, Members, Loans.

Include indexing suggestions.
```

Generate queries for:
- Issued books
- Overdue books (>30 days)
- Books count per member

- **2) Code**

```
CREATE TABLE Books (
    book_id INT PRIMARY KEY,
    title VARCHAR(50)
);

CREATE TABLE Members (
    member_id INT PRIMARY KEY,
    name VARCHAR(50)
);

CREATE TABLE Loans (
    loan_id INT PRIMARY KEY,
    book_id INT,
    member_id INT,
    loan_date DATE,
    FOREIGN KEY (book_id) REFERENCES Books(book_id),
    FOREIGN KEY (member_id) REFERENCES Members(member_id)
);

-- Queries
SELECT * FROM Loans;

SELECT * FROM Loans
WHERE DATEDIFF(CURDATE(), loan_date) > 30;

SELECT member_id, COUNT(book_id)
FROM Loans
GROUP BY member_id;
```

- **3) Output**

Issued books listed
Overdue books identified
Loan count per member calculated

🔹 4) Remarks

✓ Indexing on IDs improves speed
✓ Date functions used for overdue check
✓ Efficient grouping queries
✓ AI improved query clarity

🔹 Task 4: Query Optimization

🔹 1) Prompt

```
Analyze and optimize SQL queries.

Convert subqueries into joins for better performance.
Explain difference in efficiency.
```

🔹 2) Code

```
-- Original
SELECT * FROM Books
WHERE author_id IN (
    SELECT author_id FROM Authors WHERE country='UK'
);

-- Optimized
SELECT b.*
FROM Books b
JOIN Authors a ON b.author_id = a.author_id
WHERE a.country = 'UK';
```

🔹 3) Output

Both queries return same result
Optimized query performs faster

🔹 4) Remarks

✓ JOIN reduces subquery overhead
✓ Better for large datasets
✓ Faster execution time
✓ AI suggested optimization

🔹 Task 5: University Course Registration

🔹 1) Prompt

```
Design a database for course registration with tables:
Students, Courses, Faculty, Registrations.

Define relationships and generate queries for:
- Students in a course
- Faculty teaching >2 courses
- Most popular course
```

🔹 2) Code

```
CREATE TABLE Students (
    student_id INT PRIMARY KEY,
    name VARCHAR(50)
);

CREATE TABLE Faculty (
    faculty_id INT PRIMARY KEY,
    name VARCHAR(50)
);

CREATE TABLE Courses (
    course_id INT PRIMARY KEY,
    name VARCHAR(50),
    faculty_id INT,
```

```
    FOREIGN KEY (faculty_id) REFERENCES Faculty(faculty_id)
);

CREATE TABLE Registrations (
    student_id INT,
    course_id INT,
    FOREIGN KEY (student_id) REFERENCES Students(student_id),
    FOREIGN KEY (course_id) REFERENCES Courses(course_id)
);

-- Queries
SELECT s.name
FROM Students s
JOIN Registrations r ON s.student_id = r.student_id
WHERE r.course_id = 101;

SELECT faculty_id, COUNT(course_id)
FROM Courses
GROUP BY faculty_id
HAVING COUNT(course_id) > 2;

SELECT course_id, COUNT(student_id) AS total
FROM Registrations
GROUP BY course_id
ORDER BY total DESC LIMIT 1;
```

🔹 3) Output

Students listed for course
Faculty teaching more than 2 courses identified
Most popular course retrieved

🔹 4) Remarks

✓ Proper relational mapping
✓ Many-to-many handled via Registrations
✓ Efficient joins and grouping
✓ AI improved schema design

Final Conclusion

✓ All schemas are normalized and structured
✓ SQL queries include joins, grouping, and conditions
✓ Query optimization improves performance
✓ AI-assisted design ensures correctness and efficiency
✓ Suitable for real-world database applications